



Fig. 2: Empty backplane of an individual PXI chassis, next to a stack of three chassis that are synchronised together

all NI data acquisition (DAQ) devices provide access to programmable function input (PFI) lines that can be used to route clocks and triggers between different devices, and software support in NI-DAQmx to easily configure these lines. Certain buses, however, have additional timing and triggering lines built-in to make multi-device synchronisation as easy as possible. PCI and PCI Express boards offer the real-time system integration (RTSI) bus, on which multiple boards in a desktop system can be cabled directly together inside the case. This removes the need for additional wiring through the front connector and simplifies I/O connectivity.

How portable should this system be?

The dramatic adoption of portable computing is undeniable and has offered engineers and scientists new ways to innovate with PC-based data acquisition. Portability is an important factor for many applications. It could easily be the primary reason to choose one bus over another. In-vehicle data acquisition applications, for example, benefit from hardware that is compact and easy to transport. External buses like USB and Ethernet are particularly good for portable data acquisition systems because of quick hardware installation and compatibility with laptop computers.

Bus-powered USB devices offer additional convenience because they do not require a separate power supply and are directly powered through the laptop's USB port. Using wireless data transfer buses is another good option for portability because the measurement hardware itself becomes portable while the computer can remain stationary.

At the Indian Institute of Technology (IIT) Madras, Professor Krishnan Balasubramaniam developed a pipe inspection system for the oil and gas industry that detects corrosion using ultrasonic waves. He needed a system that was portable, to travel along the length of pipelines and perform analysis on the pipe-integrity. Using a laptop with LabVIEW software and USB-based measurements, he was able to develop a system that could detect pinhole defects, and could potentially avoid unexpected down time.

How far will the measurements be from my computer?

The distance between where the measurements are to be carried out and the computer's location can drastically vary from application to application. To achieve the best signal integrity and measurement accuracy, you should place your data acquisition hardware as close to the signal source as possible.



Fig. 3: USB bus-powered data acquisition device with direct BNC connectivity



Fig. 4: Wireless measurement system

This can be a challenge for large distributed measurements like those used for structural health monitoring or environmental monitoring. Running long cables across a bridge or factory-floor is costly and can result in noisy signals. One solution to this problem is to use a portable computing platform to move the entire system closer to the signal source. With wireless technology, the physical connection between the computer and the measurement hardware is removed altogether. You can take distributed measurements and send the data back to a central location.

Choose the bus that works best for you

By asking a few basic questions up front, you can easily decide on the bus or form factor that ideally suits your application. Considering factors like bus bandwidth, single-point requirements, synchronisation, portability and distributed needs can help to narrow down the list of hundreds of devices from different vendors. ●

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